COURSE
OVERVIEW

PORTABLE LADDER USER

The Portable Ladder User course is designed to provide learners with the knowledge and practical skills required to safely select, inspect, erect, use, and transport portable ladders in the workplace.

The course focuses on ladder types, limitations, safe working loads, inspection requirements, and correct climbing and working techniques. Emphasis is placed on hazard identification, maintaining stability and three points of contact, working near power lines, and applying safe work procedures to reduce the risk of falls, electrical incidents, and ladder-related injuries.

BY COMPLETING THIS COURSE, LEARNERS WILL BE ABLE TO:

- Identify hazards associated with portable ladder use
- Select the correct ladder type for the task and environment
- Inspect ladders before use and identify defects
- Erect, secure, and use ladders safely
- Apply correct climbing and working techniques
- Follow safe procedures when working near overhead power lines

INDEX

INTRODUCTION

GRAVITY VALUES

R — RELATIONSHIPS

Creating and maintaining meaningful relationships with all stakeholders of Gravity.

I — INTEGRITY

Conducting day-to-day business with integrity at all times.

S — SOLUTIONS **DRIVEN**

Being solution driven.

Q — QUALITY & **EXCELLENCE**

Striving for excellence and constant improvement.

WELCOME, LEARNER!

Working at height is inherently risky — and this includes training exercises. The best way to stay safe is by learning and practising the correct techniques. This manual will help you:

- Understand the basics of fall arrest and basic rescue
- Learn how to identify and manage height-related hazards
- Build the skills needed to work safely and confidently

Working at height means doing any job where you could fall from one level to another. This includes tasks done:

- Above the ground or floor
- Near open edges
- Near fragile surfaces
- Next to holes or gaps in the ground or floor
- Close to excavations

Note: Work at height does not include slipping or tripping hazards on the same level.

HOW TO USE THIS MANUAL?

This guide supports the Fall Arrest and Basic Rescue Course and is meant to be used with the help of a trained instructor.

At Gravity Training, all facilitators are highly trained to deliver top-quality learning experiences. The content has been designed to be clear, easy to remember, and practical for real worksites.

SITE ESTABLISHMENT

Before any work begins, the site must be safe.

Use this simple checklist

Follow the Rules: 01

Make sure you follow all client rules.

Example: If night work is not allowed, do not work at night.

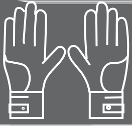
Rescue Plan 02

Have a rescue plan ready. A rescue kit and first aid kit must be on site.

Work in Pairs 03

There must be at least two workers on site.

If something goes wrong, one can help or call for help.



Check Equipment 04

All tools and gear must be safe and working before use.

Safe Site 05

Check the building or structure.

Make sure it is strong and safe to work on.

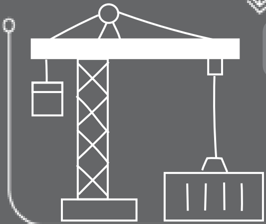
06 Manage Traffic

If there are cars or trucks nearby, there must be a traffic safety plan.

07 Do a Risk Assessment

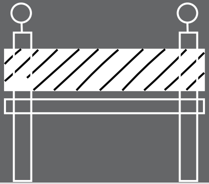
Check what dangers there are.

All workers must understand the risks before starting work.



08 Use the Right Safety Gear

Make sure the fall protection fits the job and the site.



EXCLUSION ZONES **& BARRICADES**

Exclusion Zone: An area where no one is allowed to walk or work.

Example: A 2-metre space around a hole in the roof.

Barricade: A physical barrier to block people from danger.

Example: A fence or tape to stop people from falling.

SITE SAFETY

- **Be healthy and ready** before starting work.
- **Only do jobs you are trained** and allowed to do.
- **Never work under the influence** of alcohol or drugs.
- **Always wear your safety gear** (PPE, harness if working at height).
- **Use harnesses correctly** and stay connected at heights.
- **Never work alone** and always follow safety steps.

GRAVITY

SITE SAFETY

Keep people away from areas below when working above.

Don’t walk under lifted loads or stand in dangerous areas.

Only trained people can do electrical, underground, or confined space work.

Follow road rules. Wear helmets. No passengers on open trucks..

Don’t drive tired or distracted.

Stop unsafe work! Everyone has the right to speak up.

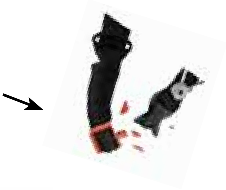
Introduction to the Full Range of Basic Personal Protective Equipment (PPE)

In order for a user to use PPE safely and correctly, there are four things each user must know about each piece of equipment, namely:

- **Identification** - The name and purpose of each piece of equipment.
- **Certification** - Only equipment with the correct South African National Standards (SANS) or European National Standards (EN) should be used and must be purchased through a reputable retailer such as Gravity Training. For example, Gravity Training only makes use of ISO 9001:2015 certified equipment suppliers.
- **Limitation** - How to use and not use equipment, the safe use criteria and hazards to identify when using each piece of equipment, and the maximum load that each piece of equipment is certified to carry. Equipment load-bearing ability is given by the MANUFACTURER as **Safe Working Load** and the **Breaking Strength** and should never be exceeded.
- **Inspection** - Pre-use inspection should be done before the start of work and that equipment should be sent to the appointed competent equipment controller for detailed inspections at least every 6 months. The use of faulty or damaged equipment may lead to serious injury or death.

- **Note that the S.W.L will differ from one manufacturer to another.**

Climbing Helmet (EN 12492)



A helmet is used to protect the technician's head during a fall as well as from falling objects:

- Must have chinstrap.
- Must be on the worker's head at all times.
- Chinstrap is designed to break under a force of 50 kg.
- Is designed in the a way that it can take top impacts and side impacts.

Construction Helmet EN 397

- Do not use an EN 397 Helmet for working at height .
- Reasons why one should not use these helmets, they are not rated for side or back impact forces.
- Should an object fall on the front of the helmet, it will lift up the back of the helmet and expose the user's head to potential injury.

Full Body Harness (EN361:2002, 358:2000)

A harness allows all other devices and systems to be connected to the worker. This forms the basis of all safety systems.

- Fall arrest systems must preferably be connected the chest D-link.
- The two buckles on the side of the harness are for work positioning systems.
- The harness must be fitted correctly with all straps tightened properly.
- Ensure that the buckles go to the correct corresponding connectors.

- S.W.L. = 140kg
- B.S. = 1400kg

Donning and duffing of a harness

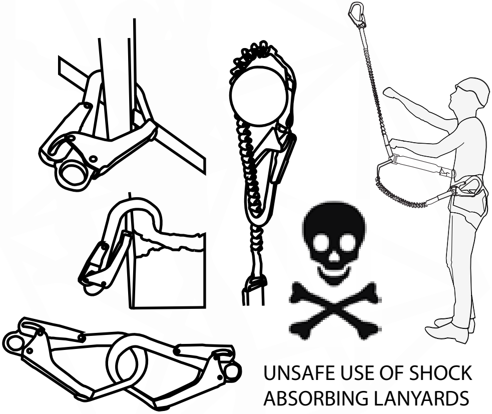
When donning a harness ensure the following:
The waistbelt is above your hips and tight.
The should straps are tight so that you can fit a fist between your chest and the harness.
The leg loops are adjusted so that they do not restrict blood flow when you crouch.
All the buckles are connected and secure.

Shock Absorbing Lanyard (EN 355:2002)

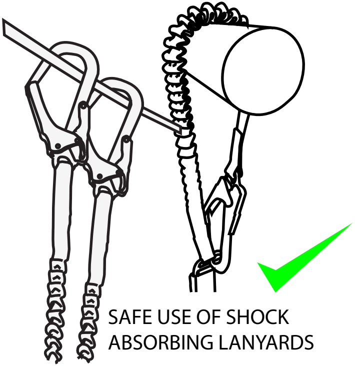
A shock absorber acts as the Primary safety when working at height.

- Shock absorber rips open (deploys) to reduce the shock load felt by the person falling.
- Must be intact during use. Once activated, the shock absorber may not be used again as it will not be able to absorb the shock load of another fall.

- S.W.L. = 130kg
- B.S. = 1300kg



UNSAFE USE OF SHOCK ABSORBING LANYARDS



SAFE USE OF SHOCK ABSORBING LANYARDS

Work Positioning Lanyard (EN 358:2000)

A work positioning lanyard is used to allow a worker to use both hands freely in order to do work effectively.

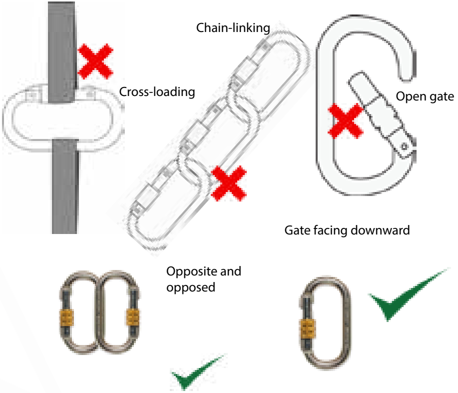
- MUST NEVER BE USED WITHOUT A FALL ARREST SYSTEM.
- Should be wrapped around the structure to prevent slipping.

- S.W.L. = 140kg
- B.S. = 1400kg

The helmet, harness, shock absorber and work positioning lanyard form the minimum PPE that must be used by a worker when at height.

Connectors (EN 362:2004)

Used to connect a range of devices and different pieces of equipment.



- S.W.L. = 220kg
- B.S. = 2200kg

Carrying Tools at Height

All tools must be tied off with tool lanyards or accessory cord to prevent tools from being dropped from height. Tools being dropped can cause damage to property and even injury or death to persons below.

Tools weighing 8 kg or less should be connected to the harness gear loops. Tools weighing in excess of 8 kg should be lifted using a separate 1:1 pulley system with a maximum lifting capacity of 20 kg. Any tools or equipment weighing more than 20 kg needs to be lifted by a qualified rope rigger. A qualified rope rigger may lift up to 40 kg with using a 1:1 system and up to 100 kg using a 3:1 system.

Pre-use Inspections

Users must inspect all the equipment to be used before work starts. This must be done to ensure that all equipment is safe for use and fit for purpose in accordance with ISO 22846-2.

Equipment is divided into 2 categories namely hardware and software, each of these have their own inspection criteria. Some pieces of equipment (e.g a harness) fall into both categories and must be inspected as such.

Software

Check for:

- Date of manufacture
- Condition of stitching
- Discolouration
- Cuts
- Burns
- Wear and tear
- Certification

Hardware

Check for:

- Rust
- Working action
- Movement
- Wear and tear
- Deformation
- Cracks

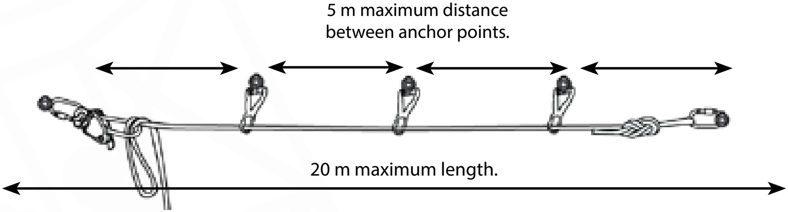
Temporary Vertical Lifelines

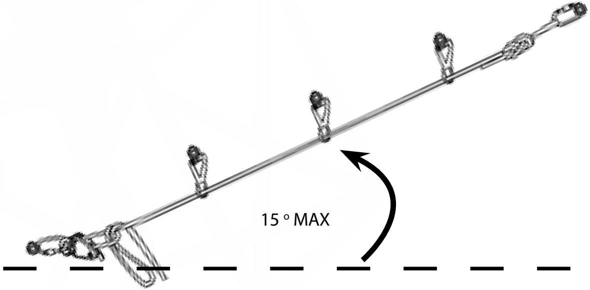
Vertical lifelines are used for easy and quick movement up and down a structure. These lifelines are used with a rope grab device which results in a reduced falling distance, only being limited to the amount of rope stretch.

- Maximum Length: 30 m
- Maximum persons allowed on the lifeline: 1 Person.
- Maximum angle between the 2 loops is 90°.

Temporary Horizontal Lifelines

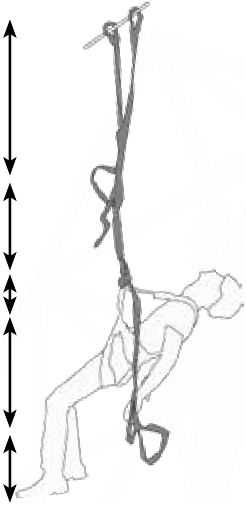
Horizontal lifelines are used for safe and quick movement from side to side. These lifelines are to be used with a shock absorber which can slide from side to side as the technician moves.





Minimum Free Space

Minimum free space (MFS) is the open space a technician would need below him to safely arrest a person's fall without hitting the ground or a structure. Reducing the fall distance or Fall Factor would reduce the required MFS.



2m for the lanyard

1.5m for the lanyard, since at Fall Factor 2 the lanyard will deploy completely

0.5m for the harness stretch

1.5m for the person's body.

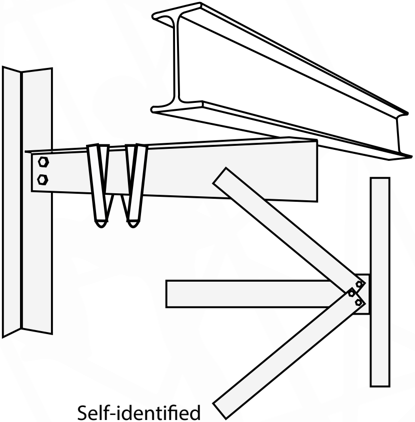
0.5m free space so that the person does not hit the ground

THUS: 2 m+1.5 m+0.5 m+1.5 m+0.5 m = 6 m

Anchor Points

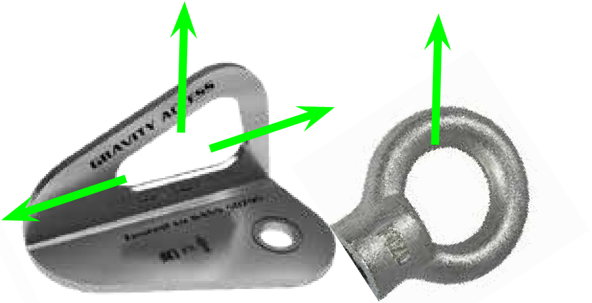
Anchor points are intended to be the last remaining connection point between a worker and a structure should a fall occur to prevent the worker from falling to the ground.
Anchor points are mainly divided into two categories:

- Self-identified anchor points, and
- Planned anchor points (EN 795:2012).



Self-identified

- S.W.L. = 150kg
- B.S. = 1500kg



Gravity Access Anchor

Eye-nut

Planned

- S.W.L. = 120kg
- B.S. = 1200kg

B. Introduction to Ladders:

General safety regulations 6:

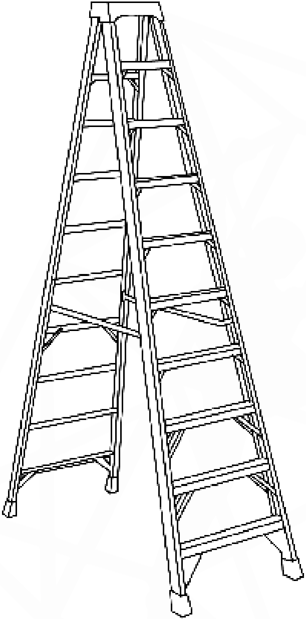
"No employer shall require or permit any person to work in an elevated position, and no person shall work in an elevated position, unless such work is performed safely from a ladder or scaffolding, or from a position where such person has been made as safe as if he were working from scaffolding."

Also see General safety regulations 13A

Due to the fact that ladders have a high risk connected to them, it is always important to consider if a ladder is suitable for the work that is to be performed.

Consider the following:

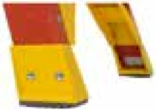
- The task that is to be performed (light work only).
- The time it will take to complete the task (no longer than 30 min).
- Are there any alternative access methods that can be used to
  accomplish the same goal. (e.g. rope access, MEWP's etc.)



Selecting a Suitable Ladder

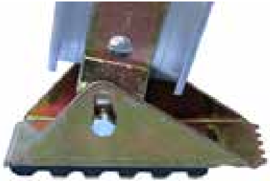
Ladder Feet

When selecting a ladder, remember to ensure that the ladder is fitted with the correct feet for the surface that is to be worked on.



1. Non-slip Feet

Used on flat surfaces such as concrete where the ladder can slip backwards.



2. Swivel Feet

Used on angled surfaces to stabilize the ladder with minimal effort.

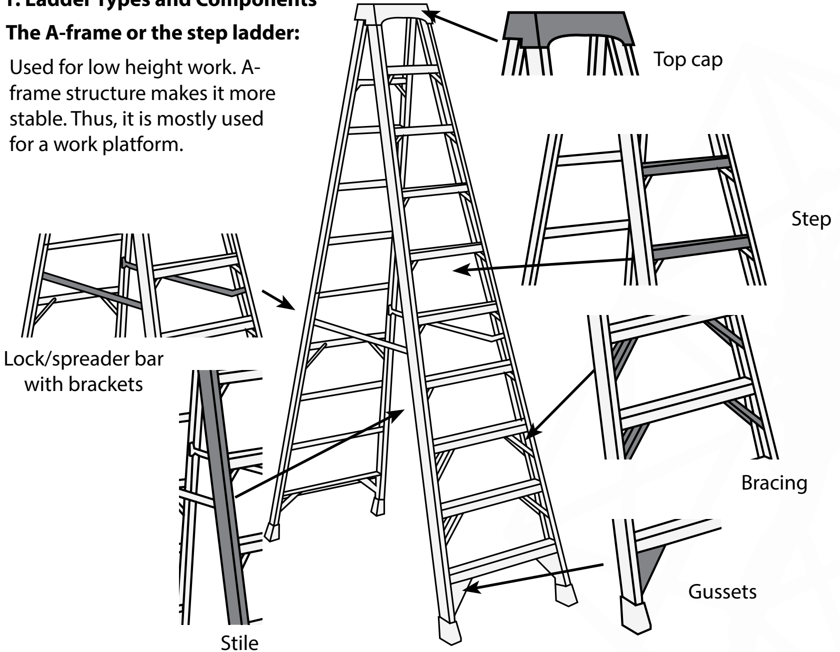
3. Spikes

Used on soft ground where the ladder can sink into the ground such as a grassy field. The spikes anchor the ladder securely in the ground preventing any unwanted movement.

1. Ladder Types and Components

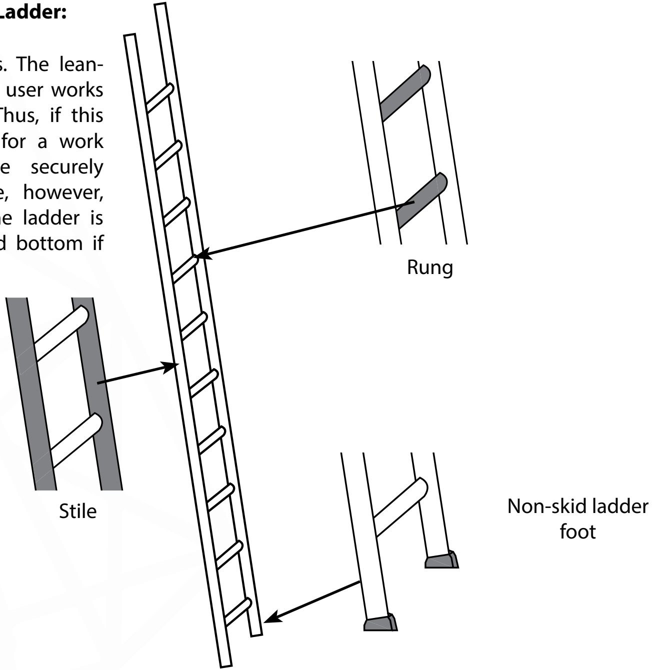
The A-frame or the step ladder:

Used for low height work. A-frame structure makes it more stable. Thus, it is mostly used for a work platform.

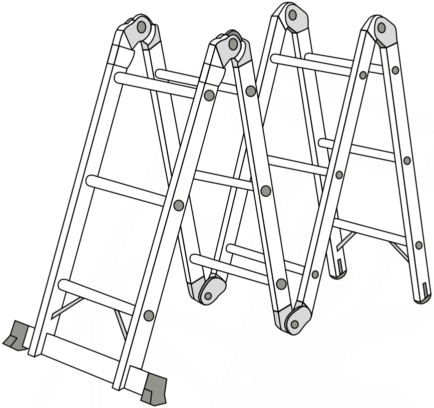


The Lean-to or Single Ladder:

Used for higher access. The lean-to ladder can tip if the user works outside of the stiles. Thus, if this ladder is to be used for a work platform, it must be securely anchored at the base, however, it is preferable that the ladder is secured at the top and bottom if possible.

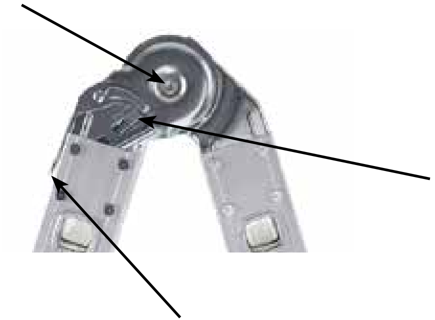


Multi-purpose or Articulated Ladder:



Multi-purpose ladders are used for a variety of scenarios and are very versatile. However, because of the hinges that give the ladder it's versatility, it is also one of the more dangerous ladders to use, thus:

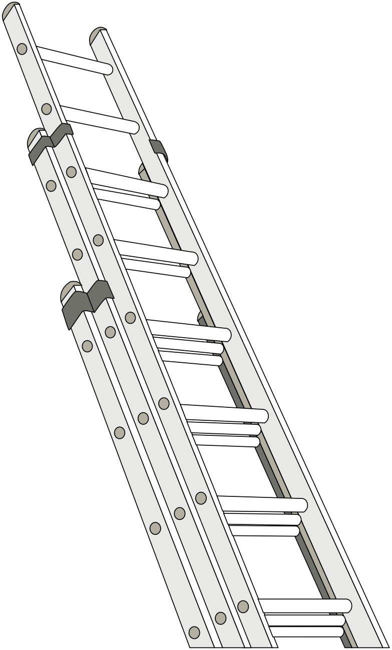
- Always check to see if the locks are properly engaged; check both physically and visually.



Extension Ladder

Extension ladders are used for high access work. The benefit of the extension ladders is that they can be used as normal lean-to ladders if needed, or they can be extended should more height be required.
However:

- Always ensure that the locks engage properly.
- Should a ladder be required to extend, two people or more are required to extend the ladder safely.



2. Height of Task

Always consider the height of the task that needs to be performed. This will determine which type of ladder will be used. For example: an A-frame ladder will not be suitable for work above a certain height, and an extension ladder may not be suitable for any work required to be done below 5 m. Connect a proximity tester to the top of the ladder when working near overhead powerlines to ensure that the ladder does not enter a high voltage area. Good practice will be to insure that any ladder must be secured at the top and bottom in some form if possible to ensure stability and safer access.

3. Safe Working Load

Before selecting a ladder for a specific purpose. You must check that the safe working load of the ladder is sufficient. Meaning that the ladder has a rating that states the maximum amount of weight that can be put onto the ladder at any given time. Ensure that this safe working load is high enough for the task and team that will be performing the specified task.

The safe working load depends heavily on the type of material from which the ladder is constructed.

C. Ladder Material

Always ensure that the correct ladder is selected for the intended task. For example, a composite or fiberglass ladder must be used when working near or with electrical hazards.

Wooden ladders are not commonly used and will not be covered in this course, but follow all the same principles.

Composite Ladders
These ladders are made from composite materials and are used when there are any electrical hazards present during work. Hence the fact that fiberglass is non-conductive, it prevents the worker from getting an electrical shock should the ladder touch or go near a live electrical source.

Although the ladder does not conduct electricity, extreme care should be used when working with electrical hazards and the safe work procedures must be followed at all time as per the company risk assessment (e.g. switching off the main power supply).

Aluminium Ladders
These ladders are the most common as they are light enough to be carried by one person (depending on length) and are strong enough to safely support a worker at height. They are, however, not as strong as steel ladders.

D. Detailed Inspection of Ladders

Responsible person for the detailed inspection

Always ensure that a competent person does the detailed inspection on a ladder with the guidance of the OEM requirements.

The detailed inspection should be recorded and records must be kept.

All portable ladders should comply with EN 131 Standards or with the relevant standards in the users country.

Detailed inspection must be done

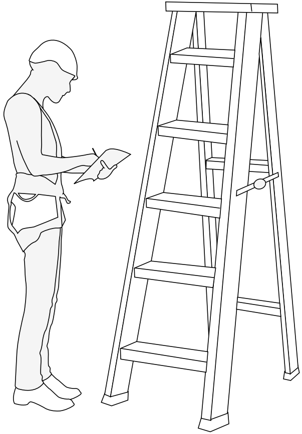
When ladders are used..	Inspect them at least every..
Daily (Frequently)	3 months
Weekly (Occasionally)	6 months
Less then monthly (Infrequently)	12 months

NOTE!!

Interim inspections may be needed in between scheduled inspections.

E. Pre-use and post-use Inspection of Ladders

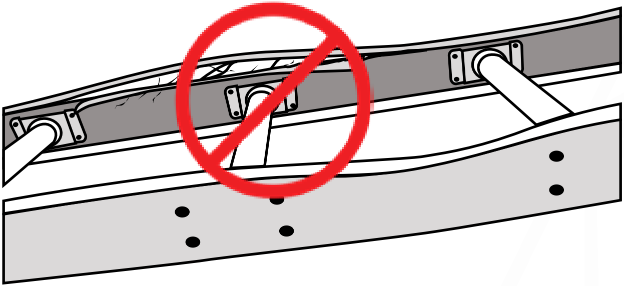
Pre-use and post-use Inspection of Ladders must be done by the user.



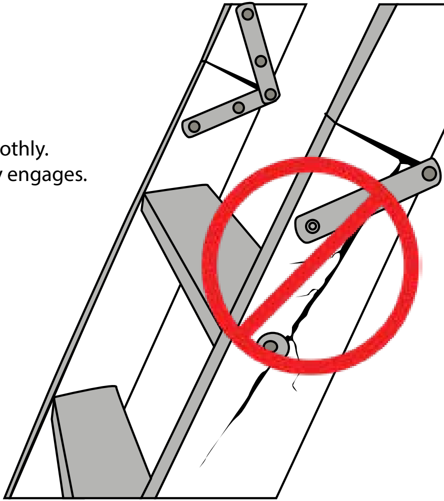
1. Always do a pre-use inspection before work starts!
2. If the ladder is dropped.
3. When left unattended.
4. if transported on a vehicle.
5. when moved from a dirty area to clean area.

6. There must be no visual defects that may affect the overall strength and stability of the ladder.

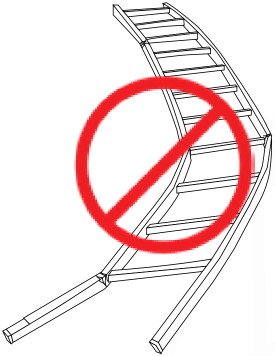
7. All basic components must be present: NO DAMAGED PARTS ARE ALLOWED!
8. Check the condition of the ropes and pulleys of extension ladders.

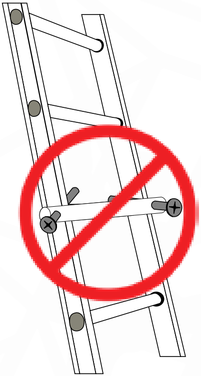


9. Ensure that all the hinges work smoothly.
10. Ensure that the locking devices fully engages.



11. Ladders should be kept clean to ensure no defects are covered by dirt and the ladder feet must be kept clean to ensure good traction on smooth surfaces.
12. Ladders must be clearly marked with a unique serial number.
13. NO LADDER WITH ANY MODIFICATION MAY BE USED.
14. Inspection registers must be kept:

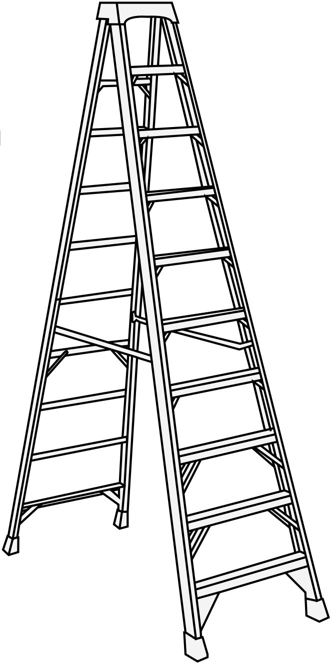




F. Erecting a Ladder

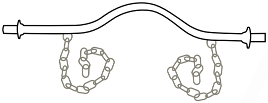
Step Ladders:

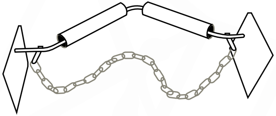
1. Always ensure that the spreader bar fully engages.
2. Ensure that the feet are stable and on firm and level ground to prevent the ladder from tilting or slipping.

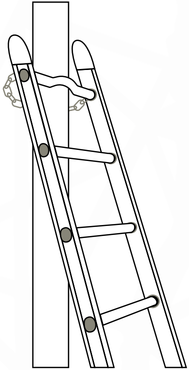


Lean-to Ladders:

1. Lift the top end of the ladder and have a worker walk away from the structure whilst "walking" his/her hands down the ladder until the top of the ladder is at the desired position, then move the base of the ladder outwards until the correct angle is reached.
2. The ladder should be secured by a worker securely holding the base of the ladder in position and/or to prevent the feet from sliding out.
3. The ladder can be secured at the top by one of two ways:
    a) By using securing bars (as shown below). These are securely attached before the ladder is elevated and either slides or runs up the pole as the ladder is lifted, this will prevent the ladder from falling backwards or sideways; or

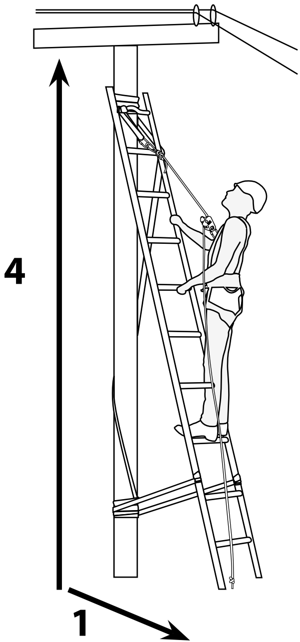






b) By securing the ladder by tying both of the stiles to the pole/structure. When using this method, a worker must secure the bottom of the ladder to allow a second worker to climb up and secure the top of the ladder without the risk of the ladder falling over. This will prevent the ladder from falling backwards. The base of the ladder can also be secured with a rope as this will prevent the base of the ladder moving backwards, causing the ladder to fall flat.

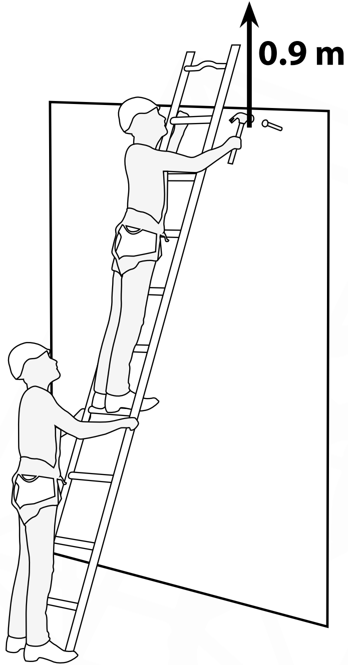
4. Always maintain a 4:1 ratio with regards to the angle. For every 4 m the ladder goes UP, the ladder must go OUT 1 m from the structure.



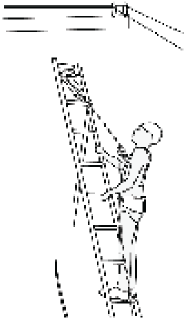
Ensure that the top of the ladder is stable against the structure.

THE TOP OF THE LADDER MUST BE 900 mm ABOVE THE WORKING AREA.

Irrespective of the securing method chosen, ALWAYS ENSURE THAT THE FEET OF THE LADDER ARE STABLE AND ON FIRM GROUND.

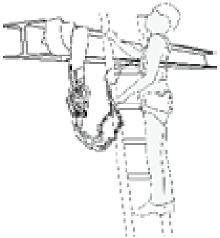


Installation of the temporary vertical life line on a pole



- Secure the ladder.
- As a first option install the remote vertical life line system (See page 24)
- If remote system can not be used, install the life line by wrapping the sling around the pole above the ladders rungs and secure the rest of the components to the sling to complete the installation.
- Connect yourself to the life line.

Creating a self identified anchor point



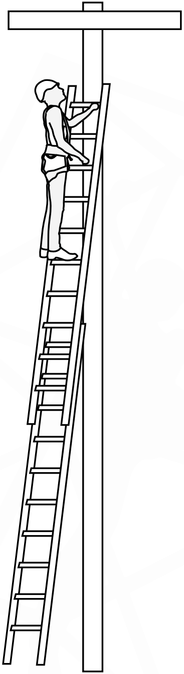
- Secure the ladder.
- Install by wrapping the sling around the structure.
- Connect your shock absorbers to the sling.

Extension Ladders:

Extension ladders are erected the same way as lean-to ladders and can be secured in the same way as well.

However:

1. Because the ladder will be extended, the angle at which the ladder rests against the structure will change. Thus, should the angle become unacceptable the base of the ladder must be moved outwards. Two workers can do this if needed.
2. Always check to see that the locks have engaged properly and are securely in place.
3. Other than the above mentioned points, the extension ladders can be approached as a lean-to ladder.
4. Extension ladders can be anchored at the top in the same way as a lean-to ladder.

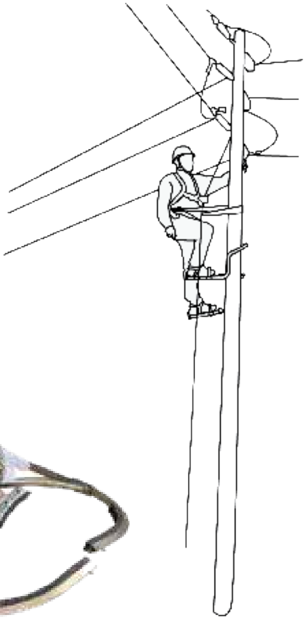


Pole Climbing Shoes:

Pole climbing shoes are an alternative to ladders and allows the user to ascend and descend wooden or concrete poles quickly and effectively.

However:

1. Pole climbing shoes must always be used with a remote vertical lifeline system as well as a work positioning system, and must be comparable with the type and size of the pole.
2. Always use a proximity detector when working near overhead power lines to avoid entering a high voltage area.
3. Always inspect the pole to be climbed (wooden, concrete, Metal and etc) before use.



4. If a remote lifeline system can not be installed (See Pg 24), Create an anchor point with a sling wrapped around the pole and connect a lifeline to the sling. This sling can then be moved up as the person climbs.

How to inspect the different type of poles

Wooden, Concrete, Metal and Composite Poles.

1. Do a visual inspection on the pole to verify if all is present, for example:
   - Surface cracks/chips
   - instability
   - Pole installation and verification tag
   - Plumpness (Perfectly Vertical)
   - Significant rebar exposure on concrete poles
   - Rust, bends and dents on metal and concrete poles
   - Rotten test on wooden poles (hit a small nail into the wooden pole)

2. Do a sound test by hitting the pole with a hammer in a GENTLE WAY to listen if the pole is save for use.

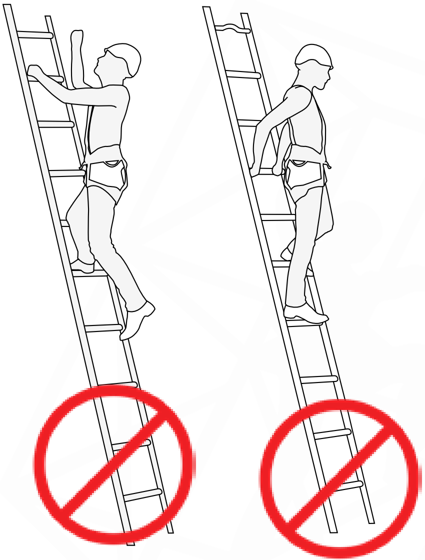
3. Ensure that the pole's diameter is compatible with the pole climbing shoes and that the right type of shoe is used on the correct type of pole.

4. If the user has any doubt, they need to refer to the manufactures specifications for the relevant pole.

Concrete pole

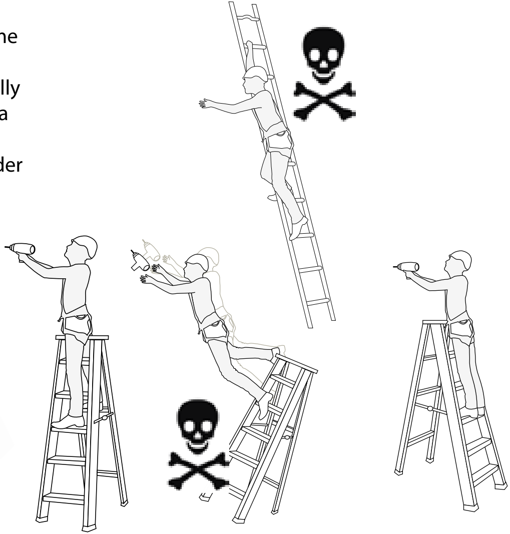
G. Ascending and Descending a Ladder

1. Always maintain three points of contact with the ladder while climbing. Maintain a firm grip with at least one hand on the ladder at all times.
2. Two contact points while climbing can result in misjudgment of stepping and instability, that greatly increases the chances of losing grip on the ladder and causing a fall.
3. Always face the ladder.
4. Never climb the on the inverted side.
5. Never have more than one person on a ladder, except if it is a double sided step ladder.
6. One step at a time, do not skip rungs.
7. Do not use the top two rungs/steps of a ladder.
8. Do not jump from a ladder.

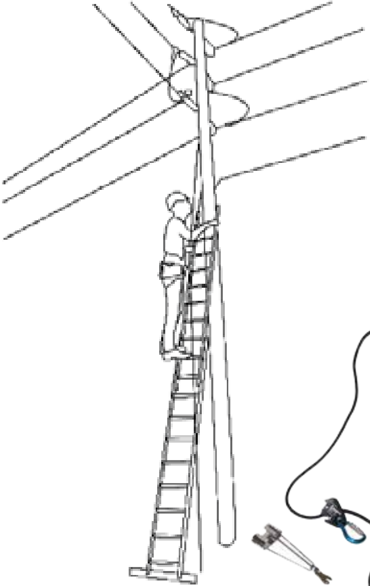


H. Working From a Ladder

1. NEVER do work in a side-on orientation from an A-frame ladder!
2. ALWAYS face the work surface head-on.
3. DO-NOT remove both hands at the same time until properly braced against the ladder.
4. Stepladders should always be used in the fully extended and locked position and never as a leaning ladder.
5. NEVER over-reach when working from a ladder to avoid toppling over.
6. ALWAYS keep your hips between the stiles!
7. Always tie tools off with tool lanyards.
8. Only carry tools that weigh less than 8 kg.



I. Using a Remote Vertical lifeline System



When using a ladder, a remote vertical lifeline system may be used to ensure the safety of the climber.

1. The remote vertical lifeline system can be connected to an anchor point from the ground.
2. This allows the 1st person to climb the ladder to be connected to the fall arrest system and therefore, the first person to climb up is already safe.
3. The system can be used with ladders or with pole climbing shoes.

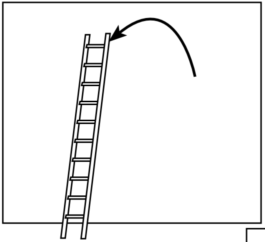
A remote vertical lifeline system consists of:

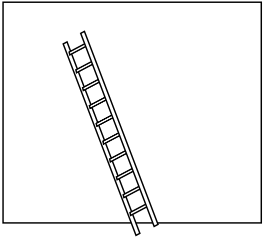
- An EN 353-2 rope grab device
- An EN 362 pylon hook
- An EN 1891 kernmantle rope
- A telescopic stick connector to connect the pylon hook
- A telescopic stick to reach the anchor point

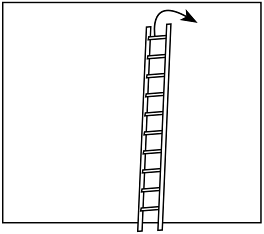
J. Moving a Ladder

When moving a ladder during work, remember the following:

1. A ladder may not be moved whilst a person is standing/working on it.
2. Step ladders must be folded up and then carried to the next position.
3. Ladders longer than 6 m must be carried by two people.
4. When moving lean-to or extension ladders, move the base of the ladder to the side first and then lift the top of the ladder away from the structure and align it with the feet. Do not attempt to move ladders more than 0.5 m at a time in this fashion.







K. Taking a Ladder Down

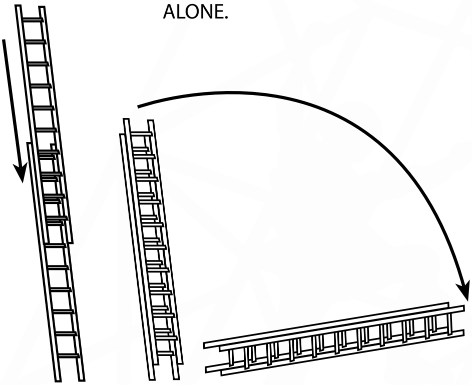
When taking a ladder down remember the following:

1. If it is an extension ladder raise the top section until the locks disengage.
2. Lower the top section until the ladder is fully collapsed. Ensure the locks are engaged.
3. Lift the ladder away from the structure.
4. Ensure that the feet of the ladder are securely anchored.
5. "Walk" ladder down until the ladder is completely grounded.

Note: A step ladder need only be collapsed and carefully lowered to the ground.

 For a lean-to ladder, follow only steps 3-5 of the taking down procedure.

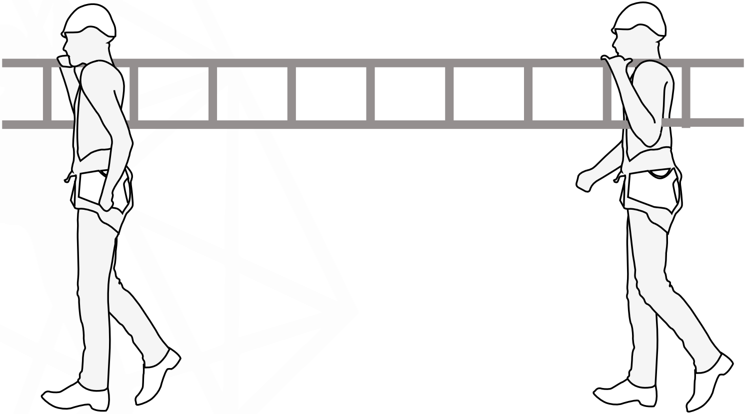
At least 2 people are required to do this. DO NOT ATTEMPT ALONE.



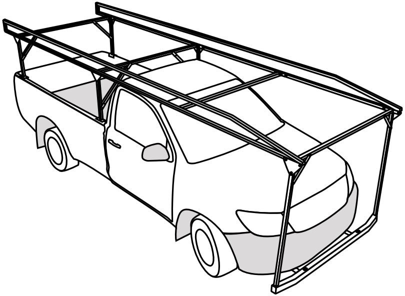
L. Carrying a Ladder

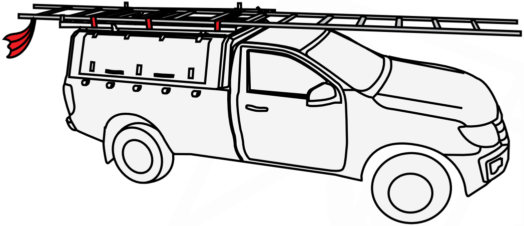
When carrying a ladder there, is a risk of the person carrying the ladder not being able to see around a corner or behind him. Therefore, extra care should be taken when transporting ladders by hand.

If the ladder is longer than 6 m or is heavy and cumbersome, it is recommended that 2 persons should carry the ladder to avoid any possibility of someone not being able to see where he/she is going or injuring people or damaging property.



M. Securing a ladder to a vehicle roof





Always ensure that the ladder is secured to the roof of the vehicle in at least two places. Making sure that the ladder does not protrude more than 1.8 m from the back of the vehicle and not more than 300 mm from the front.

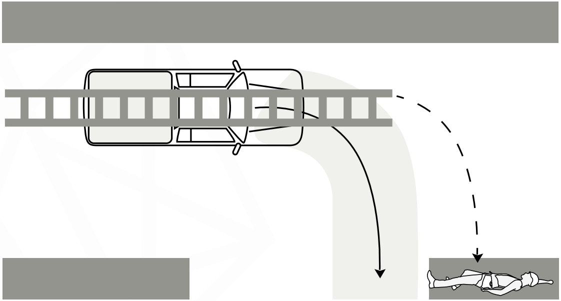
Should a ladder feel unsecured when only attached to the roof of the vehicle, a nose-cone should be used to secure the front of the ladder as well.

N. Transporting a Ladder

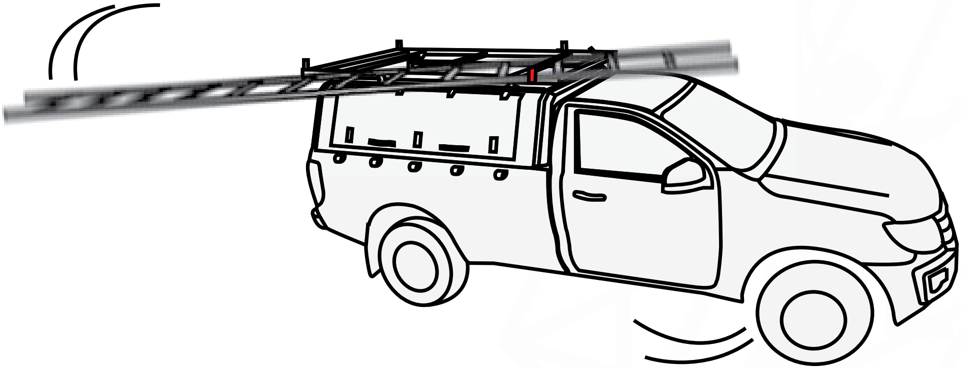
When a ladder is strapped to the roof of a vehicle it is of utmost importance that the vehicle has attached to it a secure fastening structure to prevent the ladder from sliding or coming undone or damaging the vehicle. WARNING: DO NOT OVER TIGHTEN THE SECURING STRAPS (e.g. ratchet straps) as this may cause damage to the ladder. Also ensure that the surface on which the ladder is being transported is padded.

Should the ladder extend beyond the ends if the vehicle, the ladder must be clearly marked at the ends to warn all other road users to keep a safe following distance.

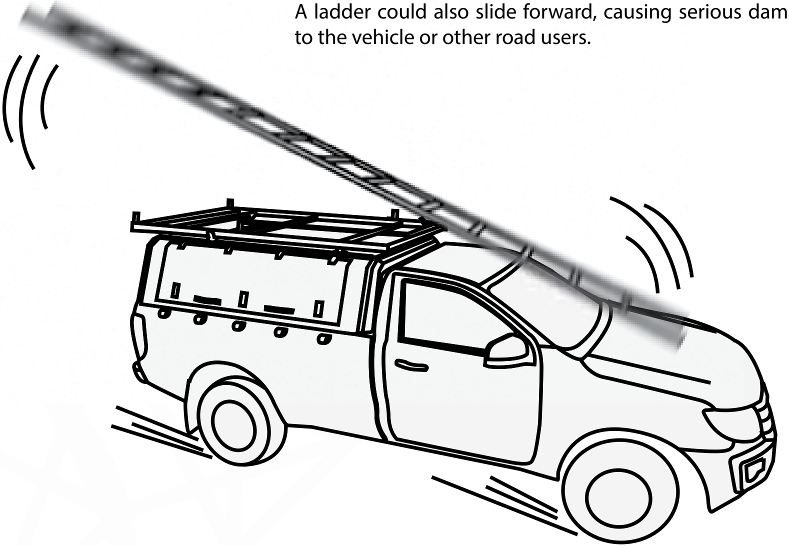
REMEMBER TO CONSIDER THE LENGTH OF THE LADDER WHEN TURING A CORNER.



Should a ladder not be secured properly, it could swing out under braking or turning which could cause serious injury to bystanders or property.



A ladder could also slide forward, causing serious damage to the vehicle or other road users.



1. Storage of Ladders

Ladders should be stored in an easily accessible area, preferably on racks with proper support along the length of the ladder at intervals of 1 m to prevent sagging.

Do not store ladders in areas with extreme temperature variations as this can weaken the ladder, or where they are exposed to the weather or to chemicals.

O. Cleaning and Maintenance of Ladders

Apart from inspecting ladders regularly, proper cleaning and maintenance of ladders will allow ladders to be used safely for longer. Ladders must be inspected and serviced by an independent ladder technician annually.

Cleaning Aluminium Ladders:

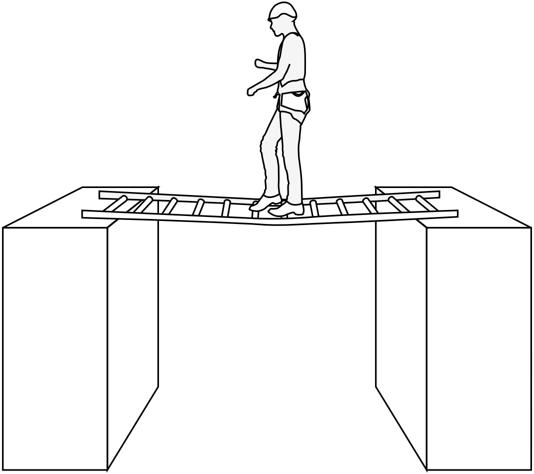
- Mild soap and water. Ensure washing of the insides of rungs and rails.
- Use ordinary car-wax to rewax all slide areas.
- Ensure that no labels are washed off; if they do come off, ensure they are replaced before use.
- Any rung replacement or serious maintenance must either be done by a qualified ladder technician or alternatively, under the direct instructions of a qualified ladder technician.

Cleaning Composite Ladders:

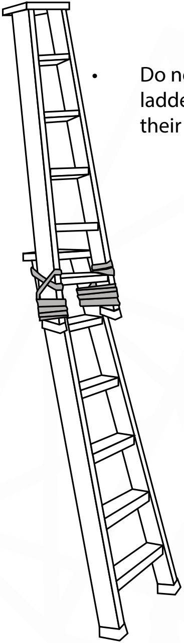
- Mild soap and water. Ensure washing of the insides of rungs and rails.
- Use ordinary car-wax to rewax all slide areas.
- Ensure that no labels are washed off; if they do come off, ensure they are replaced before use.
- Any small nicks or scratches may be sanded smooth and sealed with clear epoxy. NOTE: these types of repairs may only be used on small surface scratches as per the manufacturer's instructions. Any larger or more serious damage should be reported to a qualified ladder technician.

P. Misuse of Ladders

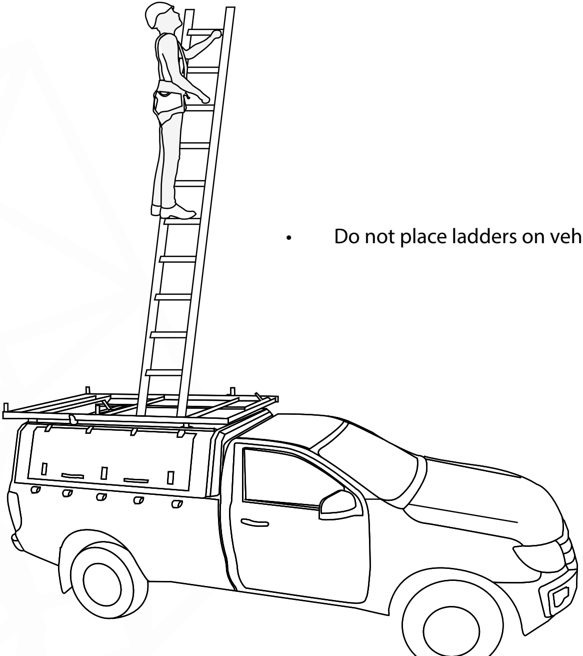
Below is a few examples of how NOT to use a range of ladders:



- Do not use ladders horizontally or as bridges.



- Do not attempt to tie ladders together to extend their reached.



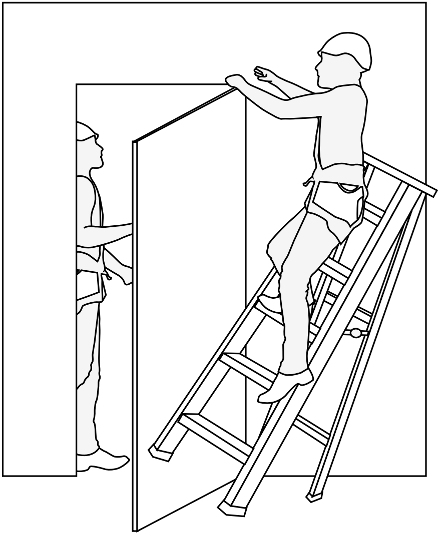
- Do not place ladders on vehicle roofs

Q. Safety in the Workplace

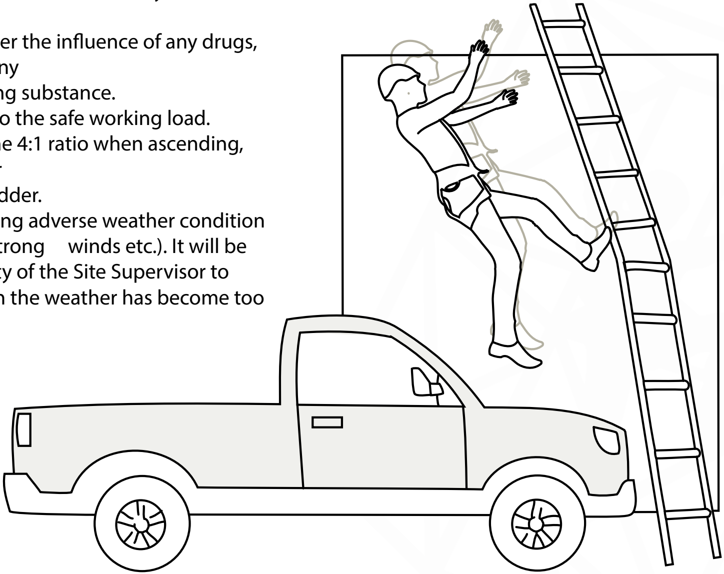
When working on ladders there are more risks and hazards than simply falling off a ladder. One must always consider:

1. Where there is any chance of moving machinery (cars, forklifts, cranes etc.) that might hit a worker or a ladder, NO WORK IS DO BE DONE UNTIL THE RISK HAS BEEN MITIGATED.
2. All work areas should be clearly marked and barricaded in order to prevent anyone from accidently entering the work area or knocking over a ladder.

3. When working around doors or entrances ensure that, either the doors are locked, or that there are clearly visible signs to inform people that there is a worker on a ladder on the other side of the door/entrance. This will prevent someone walking into a worker on a ladder.



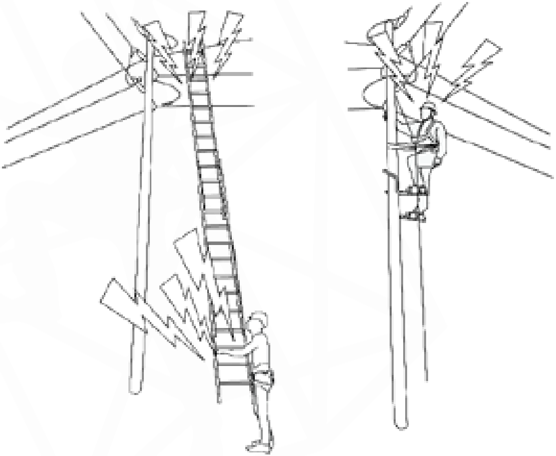
4. Always ensure that the correct PPE is used for the task, overalls,
safety boots etc. As determined by the risks assessment.
5. Never work under the influence of any drugs, medication or any
other intoxicating substance.
6. Always adhere to the safe working load.
7. Never exceed the 4:1 ratio when ascending, working from or
descending a ladder.
8. Never work during adverse weather condition (rain, thunder, strong winds etc.). It will be the responsibility of the Site Supervisor to determine when the weather has become too dangerous to
perform work.



R. Working Near Overhead Power Lines

When working near power lines the following are the main hazards that have to be guarded against:

- Electrocution - Serious injury or death caused by either direct contact or close proximity with/to a live electrical equipment
- Flashover - (also called arc flash) is the light and heat produced as part of an arc fault, short circuit or unsafe work practises.



Most electrical accidents occur when:

- Workers conduct work on equipment that was thought to be dead but which is live
- Known to be live but workers are not trained for task
- Damp or humid conditions will increase the risk of injury due the reduced effectiveness of the insulation.

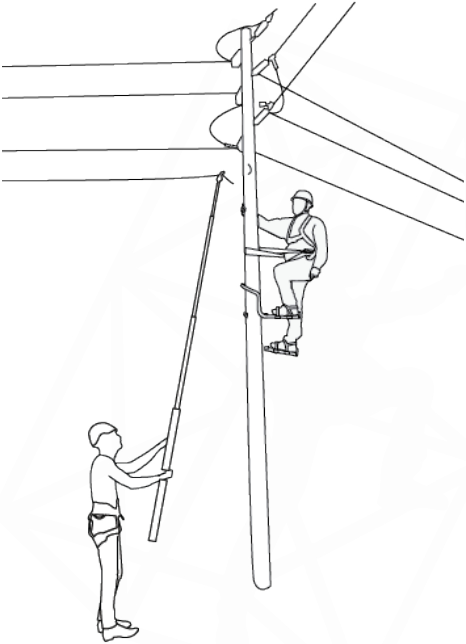
1. Live or Dead Work

As far as possible it is always preferable to conduct work on electrical equipment with the power switched OFF. However, this is not always possible, if this is the case, the following conditions must hold before conducting live work.

- It must be unreasonable in all circumstances for the conductors to be dead.
- It is reasonable in all circumstances for the person to work on live equipment.
- A full live-work risk assessment has been done
- Suitable precautions have been taken to prevent any injury.

If these conditions cannot be met then NO LIVE WORK MAY BE CONDUCTED.

DO NOT ALLOW UNAUTHORIZED OR UNTRAINED WORKERS TO CONDUCT ANY WORK ON SITE.

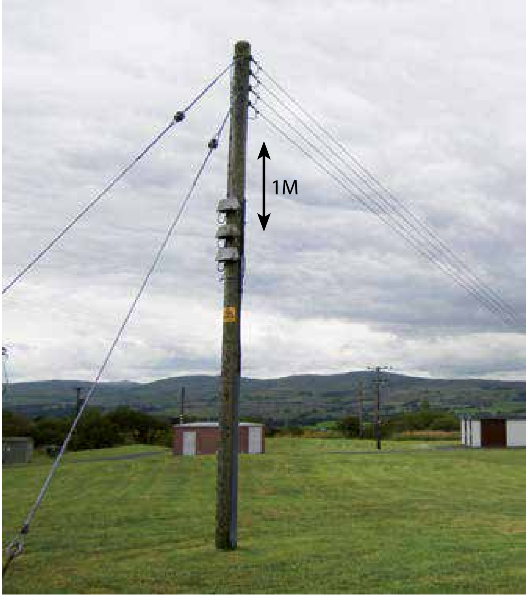


2. Types of Power Lines

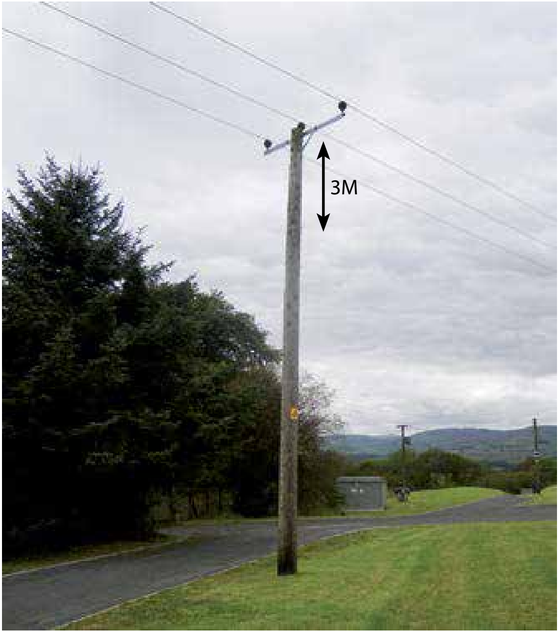
Power lines carry various strengths of voltages. Typically any voltage less than 1kV = Low Voltage, although some equipment does operate at such low voltages that they do not pose a risk of electric shock, they can still result in burns due to over-arcing.

Voltages higher than 1kV are considered High Voltage and must be treated with extreme care.

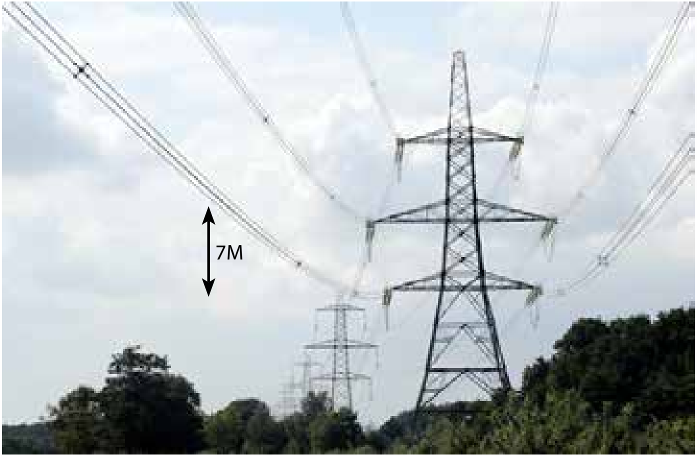
400V distribution line - ESTABLISH A **1M** EXCLUSION ZONE AROUND THE LIVE WIRES



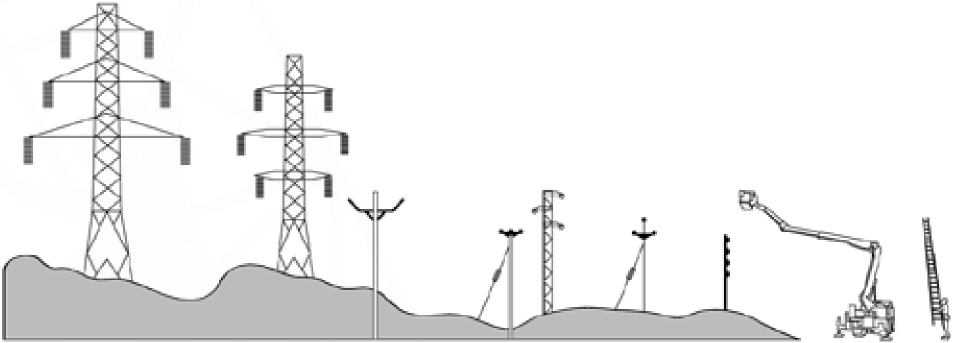
11kV - 33kV distribution lines - ESTABLISH A **3M** EXCLUSION ZONE AROUND THE LIVE WIRES



275kV transmision lines - ESTABLISH A **7M** EXCLUSION ZONE AROUND THE LIVE WIRES



All countries have legal requirements with regards to the minimum height of various types of power lines, below is an example of the most common heights

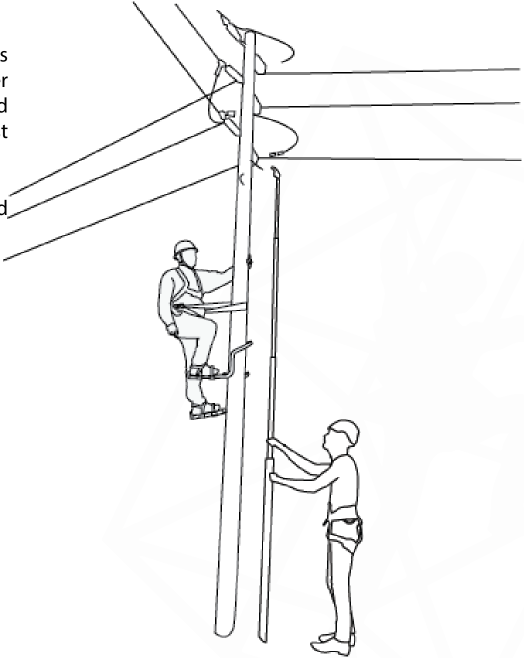


275-400 kV	132 kV	33 kV 11 kV Low voltage	
Minimum Heights	**Minimum Heights**	**Minimum Heights**	**Ensure the height of the equipment being used is less than that of the line.**
7m for 275 kV 7.3 for 400 kV	6.7m	5.2m (5.8 above roads)	

3. Measuring Heights

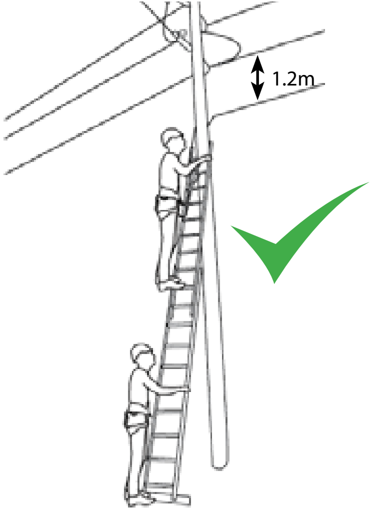
Use a telescopic stick to measure the heights of the overhead lines an determine whether they are indeed within the legal limits and if the required work can be done whilst maintaining the relevant exclusion zones.

If the exclusion zones cannot be maintained DEAD WORK MUST BE DONE.



4. Warnings

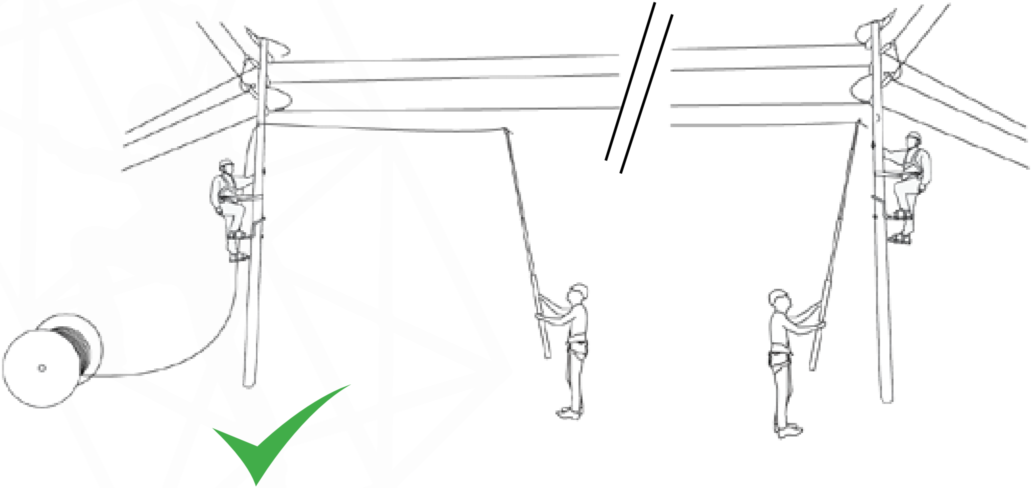
Ensure that the equipment being used NEVER enters the exclusion zones around the live wires!



Use personal alert devices at all times to ensure that workers never enter a high voltage zone!

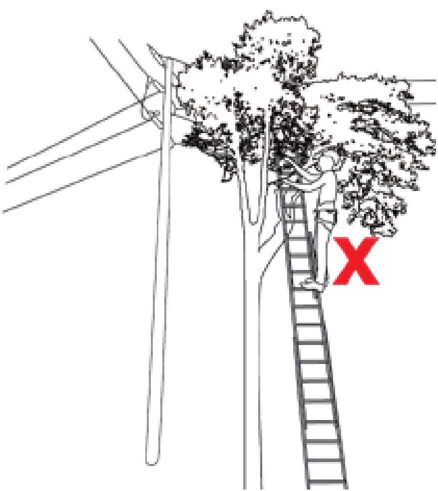
These monitors can be attached to the top of the ladder or the workers helmet and sound an alarm should either the worker or the ladder enter a high voltage area. Should the alarm sound, DO NOT ENTER FURTHER INTO THE AREA, retreat and follow the relevant procedures to ensure the safety of the workers and site.

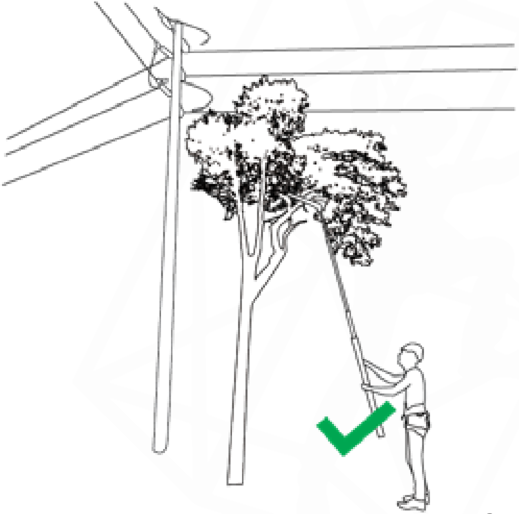
When installing any new lines on poles carrying shared route poles, ensure to keep the new lines away from the live wires at all times.



5. Tree Trimming

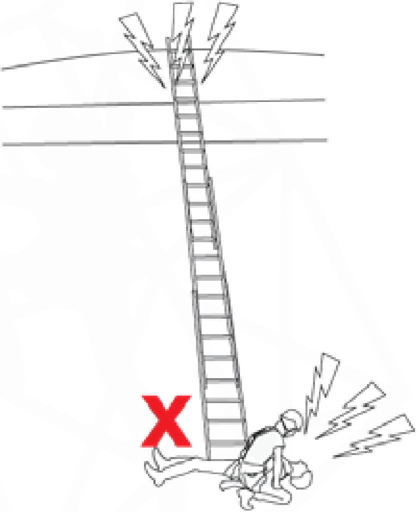
Anything such as trees obstruct the working area can conduct electricity and lead to injury or death, ensure to remove the obstruction from the ground with non-conducting equipment (e.g. fibreglass telescopic saws) before getting close to the work area.

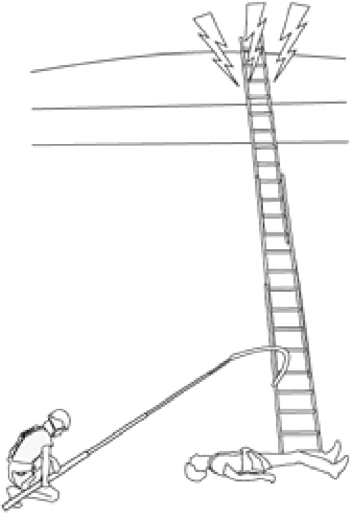




6. Emergency Procedures

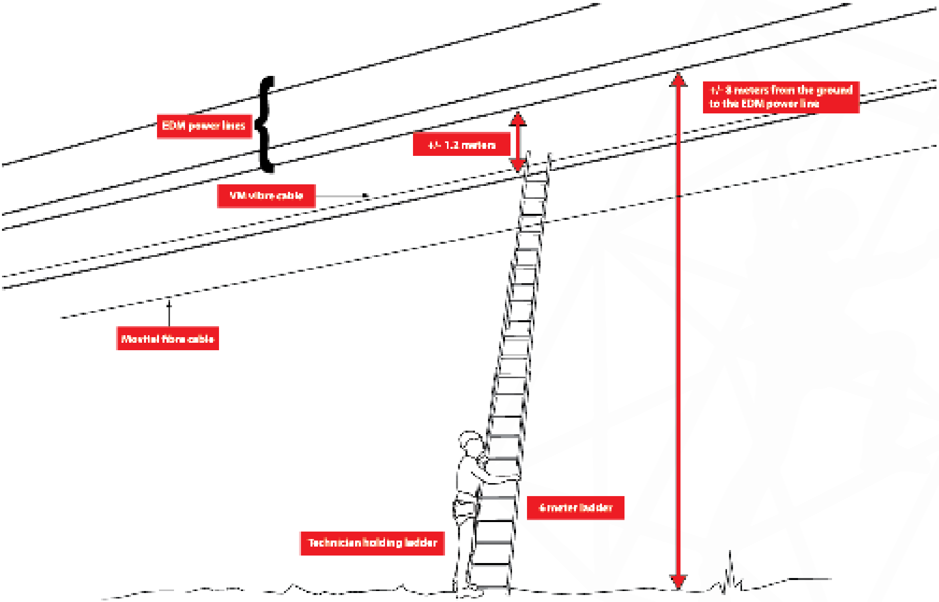
Should a worker receive a shock DO NOT touch them directly if the power has not been swithed off. If the power can not be swithced off, use a not-conducting rescue to remove the casualty from the dangerous area.





7. Example of a Typical Site

What is wrong in the picture below?



01. THE WAY FORWARD

Congratulations on completing Gravity Training's Portable Ladder User course.

If found not yet competent (NYC), you will need more training before you can be reassessed. Please do not be discouraged. Not everyone passes on their first attempt. Some learners need a bit more experience to obtain the required knowledge and skills. You will receive comprehensive feedback on your performance and be informed on how to proceed.

At Gravity, we care about the professional development of our clients. Should you wish to take the next step, consider enrolling for some of our other courses. See The Way Forward and the Gravity Skills and Development Map or consult your facilitator/assessor.